

Power BI Week 5 Project

Healthcare Dataset — Model View & Table Relationships

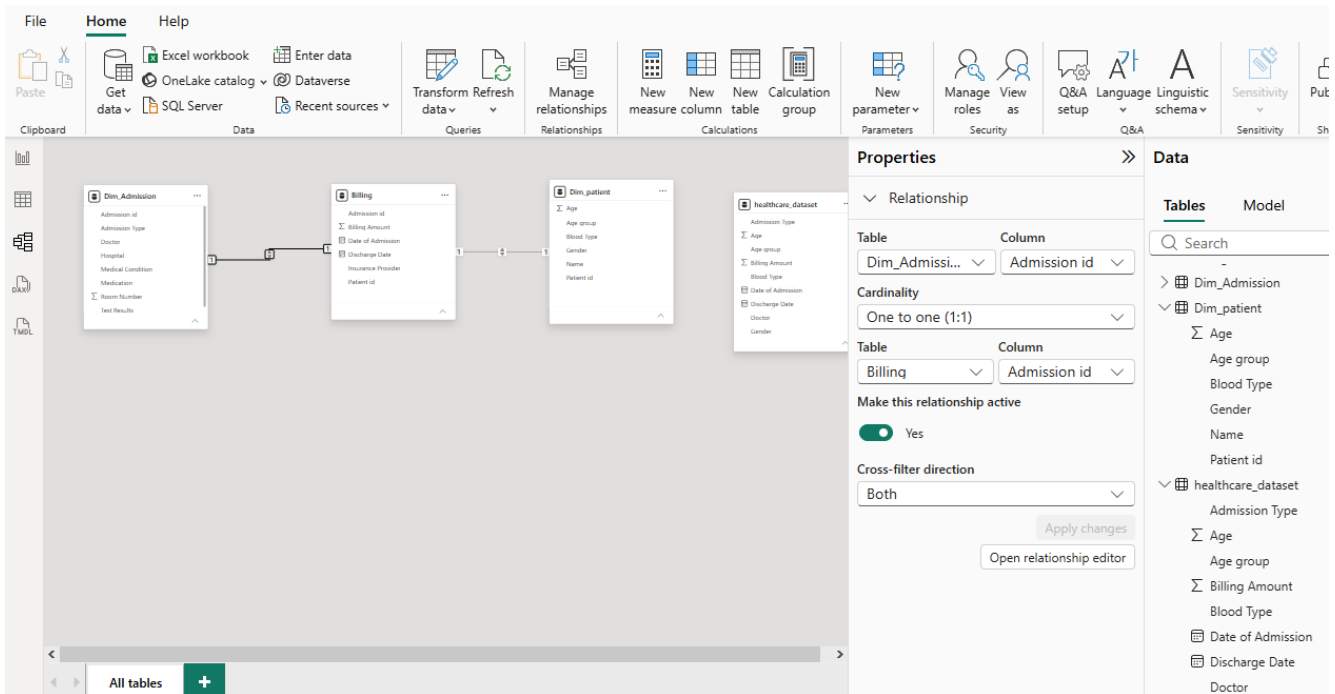


Figure 1: Model view showing the data model built for the healthcare dataset, including tables, columns, and the relationship editor for Dim_Admission ↔ Billing.

Overview

This screenshot is taken from the Model view of Power BI Desktop, where table relationships are created and managed. For this week's project, a healthcare dataset was loaded and split into logical tables to support clean, efficient reporting. The model currently consists of four tables: Dim_Admission, Billing, Dim_patient, and healthcare_dataset.

Tables in the Model

Dim_Admission stores admission-level details such as Admission id, Admission Type, Doctor, Hospital, Medical Condition, Medication, Room Number, and Test Results. This acts as a dimension table describing each hospital admission event.

Billing holds financial and stay-related information linked to an admission, including Billing Amount, Date of Admission, Discharge Date, Insurance Provider, and Patient id. This table behaves like a fact/transaction table tied back to the admission record.

Dim_patient captures patient demographic attributes — Age, Age group, Blood Type, Gender, Name, and Patient id — allowing patient details to be reused across reports without duplication.

healthcare_dataset appears to be the original imported/raw table (Admission Type, Age, Age group, Billing Amount, Blood Type, Date of Admission, Discharge Date, Doctor, Gender, etc.) that the other tables were likely derived or split from during data modeling.

Relationship Configuration

The Properties pane on the right shows the relationship currently being edited between **Dim_Admission**[Admission id] and **Billing**[Admission id]. Key settings for this relationship are:

Setting	Value
From Table / Column	Dim_Admission — Admission id
To Table / Column	Billing — Admission id
Cardinality	One to One (1:1)
Active	Yes
Cross-filter direction	Both

A one-to-one relationship means each admission record in Dim_Admission corresponds to exactly one billing record in Billing, which makes sense since one hospital admission generates one bill. Setting the cross-filter direction to **Both** allows filters to propagate in either direction between the two tables — so a slicer on a Billing field can filter Dim_Admission data, and vice versa. Similar one-to-many relationships connect Dim_patient and healthcare_dataset to the rest of the model (visible on the canvas as connector lines with 1 and * markers), enabling patient and admission details to flow correctly into visuals and measures.

Why This Matters

Structuring the healthcare data this way follows good data modeling practice: separating descriptive/dimension data (patients, admissions) from transactional data (billing) reduces redundancy and makes the model easier to query, filter, and build measures on. Correctly defined relationships and cardinalities are what allow Power BI visuals to aggregate and cross-filter data accurately across all four tables.